The Logic of the Casebook: A Synthesis

What Twenty-Four Sprachspiele Reveal About Regularity Theories of Causation

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Abstract

This synthesis distils the philosophical casebook of causal situations, intuitions, and regularity judgements assembled from Graßhoff and May (2001), Baumgartner (2008, 2013), Baumgartner and Falk (2023), and Andreas and Günther (2019, 2024a, 2024b). Read as a whole, the twenty-four cases display a strict stratification: intuitions that track Boolean structure are recovered exhaustively by non-redundancy conditions; intuitions that track the course of a concrete episode consume one further non-modal resource, instantiation; and intuitions that split structurally isomorphic pairs cannot be recovered by any theory whose analysans supervenes on structure and instantiation, forcing a principled fork between dissent and the incorporation of normality. The synthesis states this stratification, its historical logic from Mackie's hooters to structural minimality, the shared non-transitive signature of the modern theories, the exact locus of their disagreements, and the consequences for a computational implementation of causal relevance.

keywords causal regularity theory · minimal theories · causal intuitions · benchmark cases · methodology of philosophy

Table of contents

1 The method: cases as data **2**
2 The stratification result **2**
2.1 Layer one: structure and non-redundancy 3
2.2 Layer two: instantiation 3
2.3 Layer three: normality — and the impossibility that forces it 4
3 The shared signature: non-transitivity **4**
4 The epistemic root **5**
5 Consequences for implementation **5**
6 Conclusion **6**
References **6**

ℹ Note

Causal Regularity Bundle — synthesis. This document is an auxiliary component of the *Causal Regularity Bundle*. It distils the casebook (`vol3-casebook.qmd`) into its stratification result and states the consequences for any implementation of causal relevance. Foundational paper:

`arxiv-causal-regularity.qmd`; bundle concept and manifest: `bundle-hybrid-publication.qmd`, `bundle.yaml`.

1 The method: cases as data

The casebook documents a single, remarkably stable investigative practice. A situation is described in a few sentences of ordinary language; the description fixes which factors are in play and which occur; a verdict about causal relevance is elicited that competent speakers reach before any theory is consulted; and a theory is scored by whether its definitions reproduce the verdict. The practice treats these *Sprachspiele* as the empirical data of the theory of causation, in the same sense in which grammaticality judgements are the data of syntax. Three features of the corpus deserve emphasis at the outset.

First, the corpus is cumulative and adversarial. Almost every case was constructed to refute a definition: the sung song refutes unminimised sufficiency (Baumgartner 2008, sec. 4); the Manchester hooters refuted Mackie's own INUS analysis, by his own admission (Mackie 1974, 83–87); the wind-farm case refutes every regularity theory prior to 2023, including Baumgartner's own earlier definitions (Baumgartner and Falk 2023, sec. 3); the assassin's mentor refutes the deviancy condition of Andreas and Günther's 2024b theory and is answered by its 2024a revision (Andreas and Günther 2024a, fn. 2). The case method operates as forcefully inside the regularity programme as against its rivals; the theories are its products, not merely its examinees.

Second, the elicited verdicts are not uniformly stable data. For the structural cases nobody doubts the verdict: no one holds that singing lights matches or that Manchester hooters dismiss London workers. For the isomorphism cases the sources themselves disagree about what the datum is: Baumgartner reports the bogus-prevention intuition and then argues that it conflates causation with causal explanation (Baumgartner 2013, sec. 4), while Andreas and Günther treat it as a non-negotiable adequacy condition (Andreas and Günther 2024b, sec. 4.1). A synthesis must therefore keep two questions separate: whether a theory fits a given language game, and whether that language game is the one the theory is answerable to.

Third, the corpus has a generational structure that mirrors the history of the problem. The first generation (Mill and Mackie's era, systematised in Baumgartner (2008)) consists of type-level counterexamples to sufficiency and necessity: imperfect, empty, single-case, and spurious regularities. The second generation (imported from the structural-equations debate) consists of token-level episodes: overdetermination, preemption, switching, short circuits, prevention, omission. The third generation consists of theory-internal engineering: isomorphism pairs and cases like the assassin's mentor, constructed to probe exactly one clause of one definition. Each generation, it turns out, consumes exactly one new theoretical resource.

2 The stratification result

The central finding of the casebook is a stratification of causal intuitions into three layers, each matched by one layer of theory.

2.1 Layer one: structure and non-redundancy

Wherever the intuitive verdict tracks a difference in Boolean dependency structure, it is recovered — without residue and without modal resources — by the non-redundancy conditions of the minimal-theory core. The pairings are exact. Monotonically added conjuncts (the sung song) are eliminated by minimal sufficiency. Empty and contradictory antecedents (Pegasus, oxygen-and-no-oxygen) fall to the instantiation requirement, itself derivable from minimal necessity. Contraposed regularities (absent cancer, abstinent smokers) are barred by the disjunctive-normal-form syntax and the asymmetry of determination. Single-case regularities (Harold Bride's Havana cigar) fall to minimal necessity together with the minimal-complexity requirement of at least two alternative causes. Effects of a common cause (the hooters; the abstract epiphenomenon) fall to minimal necessity, the Graßhoff–May condition that Mackie's sufficiency-only minimisation lacked (Graßhoff and May 2001; Baumgartner 2008). Upstream regularities that express an exogenous factor as a redundancy-free function of its own effects (the wind-farm case) fall to structural minimality, the 2023 condition that completes the sequence (Baumgartner and Falk 2023).

The historical logic of this layer is a progressive widening of the domain over which redundancy is eliminated: Mackie minimised conjunctions; Graßhoff and May minimised disjunctions as well; Baumgartner required the minimisations to survive every suitable expansion of the factor set; and Baumgartner and Falk finally minimised at the level of whole dependency structures, eliminating biconditionals that are individually redundancy-free but dispensable within the conjunction of all biconditionals a mosaic entails. The hooters case opens this arc and the wind-farm case closes it: both are common-cause pathologies, the first solved by minimising within a regularity, the second only by minimising across regularities. The priority deserves emphasis: Mackie introduced the hooters case to show the limitations of his own INUS concept and judged the defect irreparable within regularities; the first solution — minimal necessity — is due to Graßhoff and May (2001) and, synchronously, May (1999), a quarter of a century later. The unifying principle is single: a Boolean dependency tracks causation only if every element of it is indispensable — non-redundancy (NR) — and the four conditions are its successive enforcement at the levels of conjunct, disjunct, factor set, and structure.

Two further properties of this layer matter for the synthesis. The permanence condition makes causal inference inherently diachronic and inductive: relative to an impoverished factor set the spurious hooters cluster *is* non-redundant, and only expansion reveals its redundancy, so membership in a minimal theory is warranted exactly to the degree that it has survived expansions (Baumgartner 2013; Baumgartner and Falk 2023). And the layer requires one piece of metaphysics: causal uniqueness (CU), the assumption that every complete Humean mosaic corresponds to one determinate causal structure, so that residual ambiguities (chain versus common cause; equivalent minimal theories) are always evidential, never ontological (Baumgartner and Falk 2023, sec. 5).

2.2 Layer two: instantiation

The token-level cases in which intuition still tracks structure-plus-episode — overdetermination, early preemption, late preemption — consume exactly one additional resource, and it is still non-modal: *which minimally sufficient conditions are co-instantiated in the concrete situation*. Baumgartner's active causal route requires the candidate cause to be co-instantiated with the remaining conjuncts of

its minimally sufficient condition along a chain of direct minimal theories; the preempted assassin fails this in early preemption, and the marking factor ("backup process while the effect has not yet occurred") fails it in late preemption (Baumgartner 2013, sec. 3). Andreas and Günther's maximality clause does the same work in inferential dress: the background of particular facts must be maximal, and relative to the maximal background the preempted cause no longer yields the effect (Andreas and Günther 2024b, sec. 4; 2024a, sec. 3.3). Symmetric overdetermination, the classical embarrassment of counterfactual analyses, is on this layer the trivial baseline: both disjuncts are instantiated on active routes, so both shots are causes, and no counterfactual question is ever posed. The regularity-theoretic signature of layer two is precisely this: what would have happened is never consulted; what is instantiated, and where, decides.

2.3 Layer three: normality – and the impossibility that forces it

The remaining intuitions split pairs of cases that are *isomorphic* in structure and instantiation. Bogus prevention is the isomorphic image of overdetermination; Putin's omission is structurally the neighbour's omission; the modified and isomorphic-modified extended double preventions are isomorphic by construction. This yields the casebook's one theorem-shaped fact: no theory whose analysans supervenes on Boolean structure plus instantiation can replicate all elicited verdicts, because the verdicts distinguish what the supervenience base does not. Something non-structural must enter, or the intuitions must be dissented from. That is not a defect of any particular analysis; it is a property of the data.

The fork in the literature follows deductively. Baumgartner holds the structural line: the core theory returns the isomorphism-respecting verdict (the antidote *is* a cause of survival, exactly as the second bullet is a cause of death), the divergent intuition is diagnosed as a conflation of causation with causal explanation, and a typicality-weighted, contextualised variant is offered as an explicitly pragmatic supplement whose ranking is context-relative and no part of the metaphysics (Baumgartner 2013, sec. 4). Andreas and Günther internalise the normative resource: deviancy from norms — statistical defaults or prescriptive rules such as promise-keeping — is a conjunct of the definition of causation itself, and the isomorphism pairs are split inside the theory: occurring shots are deviant and cause; normal absences of poison are not deviant and do not; Putin's default omission is no cause while the promise-breaking neighbour's is McGrath (2005). The casebook records the price of each branch. Baumgartner's branch dissents from widely shared verdicts and leaves the shape of the typicality ranking undetermined. Andreas and Günther's branch makes the deviancy clause itself a moving part: the assassin's mentor case forced its reformulation within months of publication, from root-proposition deviancy to off-path deviancy (Andreas and Günther 2024a, fn. 2), and nothing guarantees the revised clause is the last.

3 The shared signature: non-transitivity

Across the fork, the modern regularity theories share one distinctive commitment that separates them jointly from the counterfactual tradition: causal relevance is non-transitive, and the non-transitivity is not an anomaly to be managed but a direct consequence of difference-making. The switch causes the right-track passage and the right-track passage causes the arrival, but the switch makes no difference

to the arrival under any circumstances and is eliminated from every sufficient condition of it (Baumgartner 2013, sec. 3; Baumgartner and Falk 2023, sec. 6; Hall 2000). The boulder's dislodgement causes the ducking and the ducking causes the safety, but the dislodgement creates and cancels its own threat and is no cause of the safety (Hitchcock 2001; Paul and Hall 2013, 216; Andreas and Günther 2024b, sec. 4.4.1). Both theory families derive the head verdicts; Baumgartner and Falk make the point structural by locating difference-making in *atomic* minimal theories — membership in a minimal theory simpliciter does not track relevance for a specific outcome; membership in the redundancy-free biconditional *for that outcome* does (Baumgartner and Falk 2023, sec. 6).

The intra-family disagreement is confined to the tails of these very cases, and the casebook shows it to be principled. On Baumgartner's minimisation regime, once the switch and the threat factor are eliminated, the terminal outcome of the simple switch and the simple short circuit has *no* cause; Andreas and Günther recover the intuitive tail verdicts — the right-track passage causes the arrival, the ducking causes the safety — through their path condition, which licenses disregarding regularities off the candidate's path (Andreas and Günther 2024b, conclusion table). The disagreement is thus not about non-redundancy but about relativisation: whether causal judgements may be indexed to a choice of retained laws. For a notion of *difference-making relevance*, the strict verdict is the systematically motivated one; the intuitions recovered by the path condition are, on the Baumgartnerian diagnosis, intuitions about production or explanation — a diagnosis independently supported by the fact that exactly these tail intuitions are the ones expressed in transitive, processual idiom.

4 The epistemic root

The casebook also makes visible a unification that is easy to miss when the papers are read separately. The Andreas–Günther theories did not begin as theories of causation but as an analysis of explanatory 'because' by a strengthened Ramsey test: suspend judgement about antecedent and consequent, add the antecedent hypothetically, and ask whether the consequent becomes inferable (Andreas and Günther 2019). Every distinctive device of their later causal theories is prefigured there. The agnostic contraction that blocks Munich/Lund reappears as the removal of the effect's regularity in the setting operation; the belief-base asymmetry that orients tower and shadow reappears as the forward-directedness of settings over directed regularities; the contractible backgrounds that admit both overdetermining explanations of the forest fire reappear as the existentially quantified F' and L' of the causal definitions. Causation, in their final slogan, is deviant forward-directed inferability along the causal paths of direct non-redundant regularities — a Ramsey-test notion with a normality filter, grafted onto a Baumgartnerian stock of non-redundant regularities. The casebook's three layers thus also describe the composition of this theory: its regularities are layer one, its settings and maximal backgrounds layer two, its deviancy clause layer three.

5 Consequences for implementation

For a computational framework that implements causal relevance in the Graßhoff–May tradition, the synthesis yields four directives.

The structural layer is complete and mechanisable. Minimal sufficiency, minimal necessity, permanence under factor-set expansion, and structural minimality are Boolean conditions over coincidence data; the twenty-four cases fix their extension on every disputed configuration, and the type-level cases of the casebook constitute a regression suite that any implementation must pass: the hooters cluster, the sung song, the Havana cigar, and the upstream wind-farm biconditional must be eliminated; the two-soldier disjunction must survive.

Token-level queries require an instantiation layer, not a counterfactual one. An implementation extends to actual causation by tracking which minimally sufficient conditions are co-instantiated per case row — the active-route check — including marking factors where the data distinguish process stages. Nothing in the benchmark suite requires interventions or counterfactual evaluation.

Normality should be implemented as Baumgartner's contextualised variant: an optional, explicitly flagged typicality weighting over configurations, separate from the core relevance verdict. The casebook shows that any attempt to build normality into the core either dissents from the overdetermination baseline or inherits the instability documented by the assassin's mentor. Returning both verdicts — structural relevance, and context-weighted relevance where a ranking is supplied — preserves the information that the two notions can disagree, which is itself the content of the isomorphism cases.

Finally, verdicts should be reported with their deciding condition. The casebook demonstrates that "is *A* causally relevant to *E*?" is answered by different clauses in different structures, and that the clause carries the explanation: the hooters are excluded *because* minimal necessity fails, the switch *because* minimal sufficiency fails, the preempted assassin *because* the active route fails. An implementation that returns the failed condition alongside the verdict reproduces not only the intuitions but the arguments of the tradition — which is what distinguishes a theory-grounded system from an oracle.

6 Conclusion

Read as a single document, the casebook supports a precise claim: the regularity-theoretic programme has replicated every causal intuition that can be replicated by a theory of difference-making over Boolean structure and instantiation, and it has located, exactly, the residue that cannot. The residue is normative, its treatment is a genuine fork, and both branches of the fork are represented within the tradition by worked-out theories whose disagreements are confined to predictable places — the tails of non-transitive structures and the isomorphism pairs. Mackie's hooters, which once seemed to show that regularity theories confuse regular succession with causation, now marks the point where the programme began to distinguish them more finely than the intuitions themselves: the casebook's last lesson is that where structure, instantiation, and norm come apart, the theory can say so, and say which.

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